

payrollsMomentum

Technical Function Documentation

Quantitative Strategy Designer — SignalBuilder

+Model/payrollsMomentum.m

Series	FRED PAYEMS — Total Nonfarm Payrolls (thousands, SA)
Frequency	Monthly
Output range	(−1, +1)
Dependencies	Model.DataManager, Model.BaseModel
MATLAB	R2022b+ (built-in <code>sqlite</code>)

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1 Overview and Motivation

`payrollsMomentum` derives a smooth, bounded momentum signal from the U.S. Bureau of Labor Statistics' Total Nonfarm Payrolls series (FRED identifier `PAYEMS`), which is published monthly and seasonally adjusted. The signal is designed to:

1. Sit strictly inside $(-1, +1)$ at all times, with no discontinuities, so it can be consumed directly as a weight or risk-on/off multiplier.
2. Read strongly positive when payrolls are growing rapidly relative to their recent historical norm.
3. Read near zero (and sign correctly) when growth is decelerating or flat.
4. Read clearly negative when payrolls are declining.
5. Be robust to three distinct sources of distortion that afflict naïve implementations: isolated one-month anomalies (weather, strikes, seasonal revisions), regime changes in volatility (e.g. the post-COVID recovery), and catastrophic single-month shocks (e.g. April 2020: $-20,700\text{K}$, roughly 30 times the magnitude of a typical recession month).

The labour market is a lagging macroeconomic indicator, so the signal is primarily useful as a *regime filter* — confirming or challenging risk-on positioning established by leading indicators — rather than as a high-frequency timing tool.

2 Data Source

2.1 FRED PAYEMS

Series: Total Nonfarm Payrolls (All Employees, Thousands of Persons, Seasonally Adjusted, Monthly). Released by the Bureau of Labor Statistics on the first Friday of each month for the prior month.

- **Coverage:** January 1939 – present ($\sim 1,000$ observations).
- **Units:** Thousands of jobs.
- **Revisions:** The initial print is revised twice (preliminary and final), typically within two months. FRED always provides the most recently revised vintage.
- **Typical range:** Level $\approx 60,000\text{K}$ (1970) to $158,000\text{K}$ (2024). Monthly change $\Delta P \in [-21,000, +4,000]\text{K}$, with the April 2020 COVID shock being the extreme outlier.

2.2 Fetching and Caching

Data is retrieved via `Model.DataManager.fetch_fred('PAYEMS')`, which calls the FRED REST API on the first run and subsequently serves data from a local SQLite-backed `.mat` cache. All subsequent calls within a session — and across sessions until `update_data` is called — are served from cache at zero network cost.

3 Signal Construction Pipeline

Let P_t denote the raw `PAYEMS` level (thousands of jobs) at month t . The pipeline transforms P_t into a signal $S_t \in (-1, +1)$ through seven sequential steps. Two independent processing paths are maintained — a *numerator path* and a *denominator path* — which are combined in the final steps.

Step	Operation	Path	Output
1	First difference	Both	ΔP_t
2	Trailing median prefilter	Num.	$\tilde{\Delta P}_t$
3	Exponential moving average	Num.	M_t
4	Causal winsorization	Denom.	ΔP_t^w
5	Rolling standard deviation	Denom.	σ_t
6	Z-score	Merge	z_t
7	Hyperbolic tangent	Output	S_t

Figure 1: Signal construction pipeline.

3.1 Step 1 — First Difference

$$\Delta P_t = P_t - P_{t-1}, \quad t = 2, 3, \dots, n \quad (1)$$

ΔP_t is the net monthly job additions (thousands). It is the fundamental momentum quantity: positive values indicate job creation, negative values indicate destruction.

Rationale. Using the level P_t directly would conflate the secular upward trend in total employment (driven by population growth) with cyclical acceleration. First-differencing removes the stochastic trend and isolates the flow of new jobs — the signal relevant to business cycle positioning.

3.2 Step 2 — Trailing Median Prefilter (Numerator Path)

$$\tilde{\Delta P}_t = \text{median}(\Delta P_{t-W_{\text{pre}}+1}, \dots, \Delta P_t), \quad W_{\text{pre}} \geq 1 \quad (2)$$

where W_{pre} is controlled by `PrefilterWindow` (default 3). The median is computed over trailing (past) observations only, making it strictly causal.

Rationale. Monthly payroll data contains transient measurement noise that cannot be explained by genuine labour-market dynamics:

- *Weather distortions:* Unusually severe winter storms suppress construction and leisure hiring for a single month.
- *Strike effects:* A major labour action (e.g. the UAW strike of 2023) temporarily removes workers from the count.
- *Seasonal revision anomalies:* BLS benchmark revisions create apparent one-month spikes in historical data.

The median gate handles these cases with a key property: a single anomalous print in a window of three is always replaced by the median of its two neighbours. Two or more months moving in the same direction pass through unchanged. Contrast this with increasing the EMA half-life (Step 3): a longer EMA applies proportional additional lag to *all* movements — genuine trend shifts as well as noise. The median adds at most one month of lag to genuine regime changes while completely suppressing isolated prints.

Formally, for a window $\{x_1, x_2, x_3\}$ where x_1, x_3 are “normal” and x_2 is an anomalous spike:

$$\text{median}(x_1, x_2, x_3) = x_{\text{middle value}} \approx x_1 \text{ or } x_3 \quad (\text{not } x_2) \quad (3)$$

When `PrefilterWindow` = 1, Step 2 is a no-op: $\tilde{\Delta P}_t = \Delta P_t$.

3.3 Step 3 — Exponential Moving Average (Numerator Path)

$$\alpha = 1 - \exp\left(-\frac{\ln 2}{\tau}\right) \quad (4)$$

$$M_t = \alpha \tilde{\Delta P}_t + (1 - \alpha) M_{t-1} \quad (5)$$

where τ is `SmoothHalfLife` (default 3 months).

Half-life semantics. The smoothing factor α in equation (4) is derived from the half-life condition: the weight assigned to an observation τ periods in the past should equal $\frac{1}{2}$ the weight of the current observation. Expanding the EMA recursion:

$$M_t = \sum_{k=0}^{\infty} \alpha (1 - \alpha)^k \tilde{\Delta P}_{t-k} \quad (6)$$

The weight on $\tilde{\Delta P}_{t-\tau}$ is $\alpha(1 - \alpha)^\tau$. Setting this equal to half the weight on $\tilde{\Delta P}_t$ (which is α):

$$(1 - \alpha)^\tau = \frac{1}{2} \implies \tau = \frac{-\ln 2}{\ln(1 - \alpha)} \implies \alpha = 1 - e^{-\ln 2 / \tau} \quad (7)$$

This parameterisation is preferable to specifying α directly because the half-life has an intuitive macroeconomic interpretation: with $\tau = 3$, a payroll print from six months ago carries only 25% of the weight of the current print.

NaN handling. If FRED returns a gap in the series (revision hold-backs, early history), the EMA carries the previous value forward ($M_t = M_{t-1}$) rather than restarting, preserving continuity.

Rationale. The EMA acts as a second-stage smoother on top of the median prefilter. Together, the two-stage filter (median then EMA) provides stronger noise suppression than either alone with less aggregate lag than increasing either parameter independently.

3.4 Step 4 — Causal Expanding-Window Winsorization (Denominator Path)

Define the expanding percentile cap at time t :

$$Q_t = \text{prctile}(\{|\Delta P_s| : s \leq t, \Delta P_s \neq \text{NaN}\}, p_w) \quad (8)$$

where p_w is `WinsorPct` (default 95). The winsorized series is:

$$\Delta P_t^w = \text{sgn}(\Delta P_t) \cdot \min(|\Delta P_t|, Q_t) \quad (9)$$

Winsorization is applied only once at least 24 valid observations have accumulated (to ensure Q_t is estimated from a meaningful sample).

Rationale — the COVID problem. The April 2020 payroll print ($\Delta P \approx -20,700\text{K}$) is approximately 30 standard deviations from the pre-COVID conditional mean. Any rolling

standard deviation computed over a window that contains this observation is massively inflated. With `NormWindow` = 60 months, *every* rolling window from 2015 through 2025 includes April 2020, making the denominator σ_t roughly 10–15 times larger than its pre-COVID value. Post-COVID boom months of +800K – +1,000K then produce $z_t \approx 0.1$ –0.2, collapsing the signal to near zero despite being genuinely strong job growth.

Winsorizing at the 95th percentile replaces the –20,700K print with approximately –350K (the 95th percentile of $|\Delta P|$ computed over history up to April 2020), keeping σ_t in a range representative of normal-regime volatility.

Causality. Q_t uses only $\{|\Delta P_s| : s \leq t\}$. No future data is used, preserving the signal’s live-deployment integrity.

Separation of paths. The winsorization is applied *only* to the denominator path. The numerator EMA (Step 3) uses the prefiltered but unwinsorized series. This asymmetry is deliberate: we want the signal to correctly read *strongly positive* during a genuine hiring boom — capping the numerator would suppress the very reading we wish to amplify.

3.5 Step 5 — Rolling Standard Deviation (Denominator Path)

$$\sigma_t = \sqrt{\frac{1}{n_t - 1} \sum_{s=t-W+1}^t \mathbf{1}[\Delta P_s^w \neq \text{NaN}] (\Delta P_s^w - \bar{\Delta P}_t^w)^2} \quad (10)$$

where W is `NormWindow` (default 60 months), n_t is the count of non-missing observations in the window, and $\bar{\Delta P}_t^w$ is the window mean. This is the sample (Bessel-corrected) standard deviation, computed by `movstd` with a strictly trailing window.

Rationale — adaptive normalisation. Dividing the momentum estimate M_t by σ_t converts it to a dimensionless *relative* quantity: how many standard deviations of typical payroll-change volatility does the current smoothed momentum represent? This is essential for two reasons:

1. *Epoch comparability:* Payroll levels and absolute monthly changes have grown with the labour force. A change of +200K in 1965 (workforce ≈ 60 M) is proportionally larger than the same change in 2024 (workforce ≈ 160 M). Normalising by rolling σ accounts for this scaling.
2. *Regime sensitivity:* Volatility of monthly changes varies significantly across cycles. A 60-month window captures one full business cycle, providing a stable but responsive reference distribution.

A minimum of $\max(12, \lfloor W/2 \rfloor)$ valid observations is required; σ_t is set to `NaN` (and the signal is `NaN`) during the warm-up period.

3.6 Step 6 — Z-Score

$$z_t = \frac{M_t}{\sigma_t} \quad (11)$$

z_t is dimensionless and inherits the sign of M_t : positive when payrolls are growing above their recent volatility norm, negative when declining.

Cross-path interaction. Because M_t (numerator) is computed from the prefiltered series $\tilde{\Delta P}$ while σ_t (denominator) is computed from the winsorized raw series ΔP^w , the two paths serve different roles:

Path	Input	Purpose
Numerator	$\tilde{\Delta P}$ (median prefiltered)	Responsive, noise-gated trend estimate
Denominator	ΔP^w (winsorized raw)	Stable, shock-robust volatility reference

3.7 Step 7 — Hyperbolic Tangent Compression

$$\boxed{S_t = \tanh(\lambda z_t)} \quad S_t \in (-1, +1) \quad (12)$$

where λ is `TanhScale` (default 1.0).

Why tanh?

$\tanh$ is the natural choice for mapping an unbounded real-valued signal to a bounded interval. Its key properties for this application are:

1. **Boundedness:** $\tanh : \mathbb{R} \rightarrow (-1, +1)$ strictly, with no clipping discontinuities.
2. **Odd symmetry:** $\tanh(-x) = -\tanh(x)$, so positive and negative momentum are treated symmetrically.
3. **Linearity near zero:** For small $|z|$, $\tanh(\lambda z) \approx \lambda z$, giving fine-grained discrimination between modest positive growth and near-flat conditions.
4. **Soft saturation:** $\tanh(\lambda z) \rightarrow \pm 1$ as $z \rightarrow \pm \infty$, but never reaches ± 1 for any finite z . One extreme month cannot push the signal to its maximum; sustained directionality is required.
5. **Smoothness:** $\tanh$ is infinitely differentiable, producing no discontinuities that could cause numerical issues in downstream optimisers.

Derivative:

$$\frac{d}{dz} \tanh(\lambda z) = \lambda (1 - \tanh^2(\lambda z)) \quad (13)$$

The slope is maximised at $z = 0$ (value λ) and decays to zero as $|z| \rightarrow \infty$ — exactly the desired behaviour for a momentum signal that should be *sensitive to changes* when near neutral and *stable* when already extreme.

Effect of `TanhScale` λ :

$$\lambda \uparrow \Rightarrow \text{signal saturates faster (more decisive)} \quad \lambda \downarrow \Rightarrow \text{signal stays more graded near extremes} \quad (14)$$

At $\lambda = 1$, $z = 1$ maps to $\tanh(1) \approx 0.76$. This means a smoothed payroll print one historical standard deviation above the rolling mean produces a signal of about $+0.76$ — clearly positive but not at the ceiling, leaving room to strengthen further.

4 Calibration

Table 1 gives approximate signal values for representative payroll regimes under default parameters (`PreFilterWindow`= 3, `SmoothHalfLife`= 3, `NormWindow`= 60, `TanhScale`= 1.0, `WinsorPct`= 95). The rolling standard deviation of winsorized ΔP over the full post-1970 sample is approximately $\sigma \approx 130K$.

Regime	ΔP_t (K)	M_t (K)	σ_t (K)	z_t	$S_t \approx$
Post-COVID hiring boom	+900	+750	130	+5.8	+0.99
Strong expansion	+400	+360	130	+2.8	+0.99
Solid expansion	+200	+185	130	+1.4	+0.89
Moderate growth	+100	+95	130	+0.7	+0.61
Stalling (near zero)	+20	+20	130	+0.2	+0.16
Flat / no-growth	0	0	130	0.0	0.00
Mild contraction	-100	-95	130	-0.7	-0.61
Recession	-300	-270	130	-2.1	-0.97
Isolated 1-month spike	± 500	≈ 0	130	≈ 0	≈ 0.00

Table 1: Approximate signal values under default parameters. M_t shown as EMA-converged value after several months at the stated ΔP . The last row illustrates the median prefilter’s effect on a single-month anomaly.

5 Parameter Reference

Parameter	Type	Default	Description and guidance
PrefilterWindow	integer ≥ 1	3	Length W_{pre} of the trailing median window applied to ΔP before the EMA. 1 = disabled; 3 = removes isolated single-month anomalies; 5 = removes up to two consecutive anomalous months. Larger values reduce noise but increase the lag for genuine inflection points by at most $\lfloor W_{\text{pre}}/2 \rfloor$ months.
SmoothHalfLife	double > 0	3	EMA half-life τ in months. Controls how quickly the smoothed momentum M_t adapts to new prints. Lower ($\tau = 2$): faster, noisier. Higher ($\tau = 6$): slower, smoother. Increasing this is <i>not</i> a substitute for PrefilterWindow : it adds lag to genuine signals proportionally.
NormWindow	integer ≥ 12	60	Trailing window width W (months) for computing σ_t from winsorized ΔP^w . 60 months $\approx$ one business cycle: stable enough to smooth regime transitions, short enough to respond to structural shifts in payroll volatility over a decade.
TanhScale	double > 0	1.0	Scale parameter λ applied to z_t before tanh. Stretches or compresses the signal's dynamic range. $\lambda > 1$: signal saturates closer to ± 1 for moderate z — more binary/decisive. $\lambda < 1$: signal stays graded across a wider range of z — more analogue.
WinsorPct	double $\in [0, 100)$	95	Expanding-window percentile p_w used to compute the winsorization cap Q_t for the denominator path. 0 = no winsorization (original behaviour). 95 caps the top 5% of $ \Delta P $ observations cumulatively seen up to t . Increase to 97–98 for a lighter touch; decrease toward 90 for more aggressive shock-proofing of σ_t .
StartDate	string	""	Optional FRED fetch start date in 'yyyy-MM-dd' format. Empty string requests the full PAYEMS history (January 1939 – present). Note: signal warm-up requires at least NormWindow +2 valid observations before S_t is non-NaN.
dm	DataManager (required)		Initialised Model.DataManager handle. Must have init() called and FredApiKey set.

Table 2: Parameter reference for **payrollsMomentum**.

6 Tuning Guide

6.1 Reducing Noise Without Sacrificing Reactivity

- Increase `PrefilterWindow` from 3 to 5. This suppresses up to two consecutive anomalous prints while adding at most 2 months of lag to genuine inflections.
- *Avoid* compensating for noise by increasing `SmoothHalfLife` beyond 4–5: the resulting lag on genuine recessions and recoveries becomes macroeconomically significant.

6.2 Making the Signal More Decisive

- Increase `TanhScale` to 1.3–1.5. Solid expansion months (+150K, $z \approx 1.1$) then map to $\tanh(1.5 \times 1.1) \approx 0.94$ rather than 0.80.
- The signal still cannot reach ± 1 for any finite input.

6.3 Post-Shock Recovery

- If the signal is visibly muted for several years following a shock event, increase `WinsorPct` from 95 to 97. This applies a lighter cap, preserving more of the historical volatility in σ_t while still preventing the catastrophic outlier from dominating.
- Do not increase `WinsorPct` beyond 99: at that point the protection against genuine extreme shocks is lost.

6.4 Faster Reaction to Trend Shifts

- Reduce `SmoothHalfLife` from 3 to 2. With $\alpha = 1 - e^{-\ln 2/2} \approx 0.293$, a genuine multi-month payroll shift reaches 75% of its full signal strength roughly one month sooner.
- Consider also reducing `PrefilterWindow` from 3 to 1 (disabled) if the additional half-month of lag introduced by the median gate is unacceptable in your application.

7 Outputs

The function returns a `timetable` with `RowTimes` preserved from the FRED series (UTC timezone, first day of month):

Variable	Units	Range	Description
Payrolls	K jobs	[60,000, 160,000]	Raw PAYEMS level
dPayrolls	K jobs/mo	$(-\infty, +\infty)$	Raw MoM change
SmoothedMom	K jobs/mo	$(-\infty, +\infty)$	Prefiltered + EMA momentum (M_t)
Signal	—	$(-1, +1)$	Final bounded signal (S_t)

The timetable `Description` property encodes the parameters used to generate it:

```
>> tt.Properties.Description
ans = 'PAYEMS momentum | PrefilterWin=3, SmoothHL=3 mo, NormWin=60 mo, TanhScale=1, WinsorPct=95'
```

8 Dependencies and Portability

8.1 Required Files

To run `payrollsMomentum` outside the `SignalBuilder` project, copy only:

- `+Model/payrollsMomentum.m`
- `+Model/DataManager.m`
- `+Model/BaseModel.m`

8.2 MATLAB Requirements

- MATLAB R2022b or later (built-in `sqlite` function). R2021b requires the Database Toolbox for `sqlite`.
- No Statistics Toolbox: `prctile` has been replaced by the local helper `payrollsMomentum_prctile` (nearest-rank method, implemented with `sort` and index arithmetic).
- Network access on the first run for the FRED API call. Subsequent runs are served from local cache at zero network cost.

9 Usage Examples

Listing 1: Minimal usage

```
1 dm = Model.DataManager();
2 dm.init();
3 tt = Model.payrollsMomentum(dm);
```

Listing 2: Custom parameters

```
1 tt = Model.payrollsMomentum(dm, ...
2     PrefilterWindow = 5, ... % stronger noise gate
3     SmoothHalfLife = 2, ... % faster reaction
4     TanhScale = 1.3, ... % more decisive signal
5     WinsorPct = 97, ... % lighter shock cap
6     StartDate = "1990-01-01");
```

Listing 3: Compare prefiltered vs. no prefilter

```
1 tt_raw = Model.payrollsMomentum(dm, PrefilterWindow=1);
2 tt_filt = Model.payrollsMomentum(dm, PrefilterWindow=3);
3
4 figure;
5 yyaxis left
6 bar(tt_filt.Time, tt_filt.dPayrolls, 'FaceAlpha', 0.3)
7 ylabel('MoM change (K jobs)')
8
9 yyaxis right
10 plot(tt_raw.Time, tt_raw.Signal, ':', 'LineWidth', 1, ...
11      'DisplayName', 'No prefilter')
12 hold on
13 plot(tt_filt.Time, tt_filt.Signal, '-.', 'LineWidth', 2, ...
14      'DisplayName', 'PrefilterWindow=3')
15 yline(0, 'k--'); ylim([-1.1 1.1])
16 ylabel('Signal'); legend; grid on
17 title(tt_filt.Properties.Description)
```

Listing 4: Incremental cache refresh

```

1 dm.update_data('PAYEMS');           % only fetches the delta from last cache date
2 tt = Model.payrollsMomentum(dm);    % immediately uses refreshed cache

```

10 Mathematical Summary

For reference, the complete signal in closed form (suppressing NaN handling and warm-up logic):

$$\Delta P_t = P_t - P_{t-1} \quad (15)$$

$$\tilde{P}_t = \text{median}_{s \in [t-W_{\text{pre}}+1, t]}(\Delta P_s) \quad (16)$$

$$M_t = \alpha \tilde{P}_t + (1 - \alpha) M_{t-1}, \quad \alpha = 1 - e^{-\ln 2 / \tau} \quad (17)$$

$$Q_t = \text{prctile}\left(\{|\Delta P_s|\}_{s \leq t}, p_w\right) \quad (18)$$

$$\Delta P_t^w = \text{sgn}(\Delta P_t) \cdot \min(|\Delta P_t|, Q_t) \quad (19)$$

$$\sigma_t = \text{std}_{s \in [t-W+1, t]}(\Delta P_s^w) \quad (20)$$

$$z_t = \frac{M_t}{\sigma_t} \quad (21)$$

$$S_t = \tanh(\lambda z_t) \in (-1, +1) \quad (22)$$

Parameters: W_{pre} (PrefilterWindow), τ (SmoothHalfLife), W (NormWindow), λ (TanhScale), p_w (WinsorPct).